LM Physics of Climate

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6 Radiative transfer solutions

6.1 Simple solutions of the radiative transfer equation

The radiative transfer equation (Equation 5.10) can be solved to find how the radiation field evolves as it travels through an atmosphere. The differential equation can be solved using an integrating factor:

$$\begin{aligned}
\mu_\theta \frac{\mathrm{d}I_\nu}{\mathrm{d}\tau} &= B_\nu(T) - I_\nu \\
\frac{\mathrm{d}I_\nu}{\mathrm{d}\tau} e^{\tau/\mu_\theta} &= \frac{1}{\mu_\theta}(B_\nu(T)e^{\tau/\mu_\theta} - I_\nu e^{\tau/\mu_\theta}) \\
\left[I_\nu e^{\tau/\mu_\theta}\right]_{\tau_1}^{\tau_2} &= I_{\nu,2}e^{\tau_2/\mu_\theta} - I_{\nu,1}e^{\tau_1/\mu_\theta} \\
&= \int_{\tau_1}^{\tau_2} B_\nu e^{\tau'/\mu_\theta}\mathrm{d}\tau' \\
\therefore I_{\nu,2} &= I_{\nu,1}e^{-(\tau_2-\tau_1)/\mu_\theta} + \int_{\tau_1}^{\tau_2} B_\nu e^{-(\tau_2-\tau')/\mu_\theta}\mathrm{d}\tau'. \qquad (6.1)
\end{aligned}$$

The first term in this solution represents the attenuation of the incoming radiation due to absorption. The second term represents the radiation emitted by the gas, which itself is attenuated by the remaining path of gas.

Let's consider this solution in a few end-member cases:

1. **No thermal emission from the gas:** in this case, we can set $B_\nu = 0$, which leaves $I_{\nu,2} = I_{\nu,1}e^{-\delta\tau/\mu_\theta}$, where $\delta\tau = \tau_2 - \tau_1$ is the change in optical depth through a slab of gas (Figure 1).

2. **The gas is isothermal (B_ν = constant), and $\delta\tau \gg 1$:** in this case, we can take B_ν out of the integral and evaluate it analytically. Let's set $\mu_\theta = 1$ for simplicity. Equation 6.1 becomes

$$I_{\nu,2} = I_{\nu,1}e^{-\delta\tau} + B_\nu(1 - e^{-\delta\tau}) = B_\nu + e^{-\delta\tau}(I_{\nu,1} - B_\nu). \qquad (6.2)$$

   In the limit of $\delta\tau \gg 1$, this reduces to $I_{\nu,2} = B_\nu$. That is, for very optically thick materials, we see only the thermal emission from the material and the incoming radiation ($I_{\nu,1}$ in Figure 1) is completely attenuated. This limit applies in the deeper parts of optically thick atmospheres (e.g. deep in Jupiter's atmosphere).

3. **The gas is isothermal (B_ν = constant), and $\delta\tau \ll 1$:** from Equation 6.2, $I_{\nu,2} = I_{\nu,1}$. That is, in very optically thin materials, the incoming radiation is not altered.

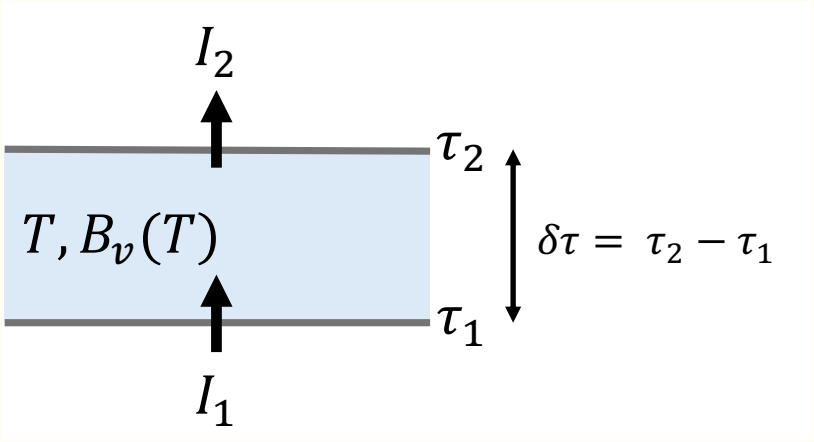


Figure 1: Radiation passes through a slab of material of temperature T with optical depth $\delta\tau$. The particles in the material emit thermal radiation proportional to $B_\nu(\tau)$.

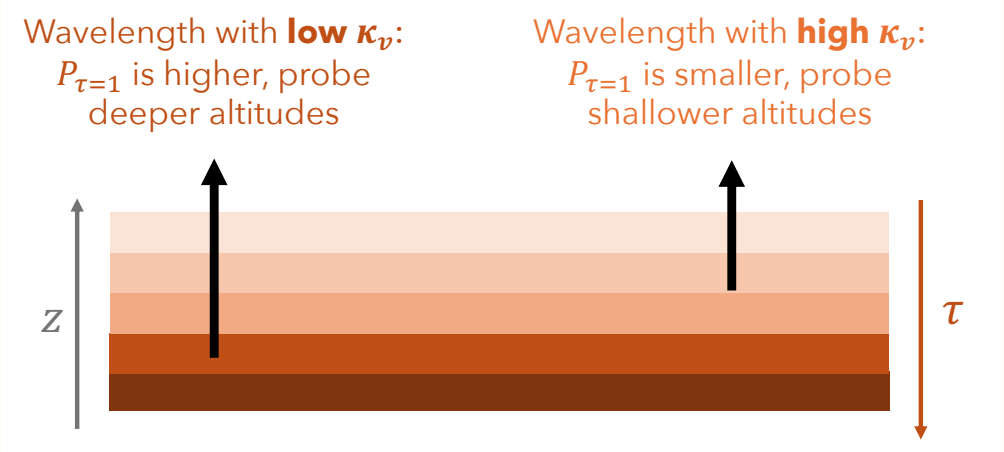


Figure 2: When looking down into an atmosphere, radiation probes different altitudes at different wavelengths. At wavelengths where the atmosphere is more opaque, radiation probes less deep into the atmosphere. Note that in this figure, we use the convention of τ increasing downwards (unlike earlier sections in these notes).

6.2 Where does the atmosphere become opaque?

Imagine we are floating in space, looking at a planet from outside its atmosphere. How deep into that atmosphere can we see? The answer is different for each wavelength of light, as the atmosphere may be more or less opaque for each, depending on the absorption cross sections of its gases (and/or aerosol particles such as clouds).

At a particular wavelength, the deepest part of the atmosphere that can be seen is referred to as the 'photosphere'. This is where the atmosphere becomes opaque for that wavelength. Another way of saying this is that as radiation enters the atmosphere, it will most likely be absorbed by the time it reaches the photosphere.

This is useful to quantify, as it can tell us how deep radiation of different wavelengths can reach. While visible light from the Sun makes it all the way to the surface, most of the high-energy UV light received by the Earth is (thankfully) absorbed in the stratosphere.

For this calculation, we will use the 'Astronomy convention' for optical depth, i.e.

$$\mathrm{d}\tau_\nu = -\kappa_\nu \rho \mathrm{d}z. \tag{6.3}$$

This applies to cases where we are looking downwards through an atmosphere. Combining Equation 6.3 with that of hydrostatic equilibrium ($\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho g$), and assuming that κ_ν is constant with pressure, we get

$$\tau_\nu = \int_\infty^z -\kappa_\nu \rho \mathrm{d}z' = \int_0^P \kappa_\nu \frac{dP'}{g} = \frac{\kappa_\nu P}{g}.$$

This means that when $\tau_\nu = 1$ (in the photosphere),

$$P_{\tau=1} = \frac{g}{\kappa_\nu}. \tag{6.4}$$

In a real atmosphere, κ_ν is not necessarily uniform, but equation 6.4 can nevertheless provide useful insights.

Note that for a mixture of gases, κ_ν is the mass-weighted sum of the absorption cross sections of the individual gases:

$$\kappa_\nu = \sum_i \frac{\rho_i}{\rho} \kappa_{\nu,i}. \tag{6.5}$$

Similarly, the total absorption cross section, σ_ν is the sum of the individual $\sigma_{\nu,i}$ weighted by number density:

$$\sigma\nu = \sum_i \frac{n_i}{n} \sigma_{\nu,i}. \tag{6.6}$$

6.3 The two-stream approximation

In the plane-parallel geometry, it is useful to consider the total radiative flux travelling vertically upwards or downwards. We can do this by integrating over the upwards/-downwards hemisphere, respectively (similar to when we calculated surface flux):

$$F_+ = \int_{\Omega_+} I_\nu \cos(\theta) \mathrm{d}\Omega$$
$$F_- = \int_{\Omega_-} I_\nu \cos(\theta) \mathrm{d}\Omega.$$

Assuming that I_ν is isotropic, we can remove it from the integral, giving:

$$\begin{aligned} F_+ &= \int_0^{2\pi} \int_0^{\pi/2} I_\nu \cos(\theta) \sin(\theta) \mathrm{d}\theta \mathrm{d}\phi \\ &= \pi I_\nu. \end{aligned}$$

Similarly,

$$\begin{aligned} F_- &= -\int_0^{2\pi}\int_{\pi/2}^{\pi} I_\nu \cos(\theta)\sin(\theta)\mathrm{d}\theta\mathrm{d}\phi \\ &= \pi I_\nu. \end{aligned}$$

We can do these integrals for the entire radiative transfer equation. In the upward direction,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\tau}\int_{\Omega_+} I_\nu \cos(\theta)\mathrm{d}\Omega &= \int_0^{2\pi}\int_0^{\pi/2} B_\nu(T)\sin(\theta)\mathrm{d}\theta\mathrm{d}\phi - \int_0^{2\pi}\int_0^{\pi/2} I_\nu \sin(\theta)\mathrm{d}\theta\mathrm{d}\phi \\ &= 2\pi B_\nu(T) - 2\pi I_\nu, \end{aligned}$$

resulting in

$$\frac{1}{2}\frac{\mathrm{d}F_+}{\mathrm{d}\tau} = -F_+ + \pi B_\nu(T). \tag{6.7}$$

Similarly,

$$\frac{1}{2}\frac{\mathrm{d}F_-}{\mathrm{d}\tau} = F_- - \pi B_\nu(T). \tag{6.8}$$

Note that the average value of μ_θ, $\overline{\mu_\theta} = \overline{\cos(\theta)} = 1/2$. So, the average path length in optical depth units is $\overline{\tau_\nu} = \tau_\nu/\overline{\mu_\theta}$ (see Figure 4 in Section 5). Note that $\overline{\tau_\nu} > \tau_\nu$ as most photons travel at an angle to the vertical and therefore travel further.

This now allows us to rewrite Equations 6.7 and 6.8 as:

$$\frac{\mathrm{d}F_+}{\mathrm{d}\overline{\tau}} = -F_+ + \pi B_\nu(T) \tag{6.9}$$

and

$$\frac{\mathrm{d}F_-}{\mathrm{d}\overline{\tau}} = F_- - \pi B_\nu(T). \tag{6.10}$$

These are known as the *two stream equations*, and they will be very useful when calculating radiative energy transport.